Risk-Budgeted Mean-Variance Portfolios

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Intro

- Portfolio allocation is a central question in Finance;
- Two major issues from the practitioner perspective:
    1. How to model the stochastic process driving returns?
    2. Given a stochastic process for returns, how to allocate money?

This paper is about the latter, not the former.

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The cornerstone: Mean-Variance (MV)... but nothing comes for free;

Mean-Variance (MV)

- Best trade-off between risk (variance) and expected returns;
- Leads to concentrated portfolios;
- Not very robust to moment estimation error;

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- Explicit guard against portfolio concentration;
- No control over expected returns;

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This paper: Risk-Budgeted Mean-Variance (RBMV) portfolio;

- RBMV nests both the RB and MV frameworks;
- Makes explicit the tension between expected returns and risk concentration;
- Measures how much concentration you need to accept to get higher returns;

Literature

Mean-Variance

- Markowitz (1952)
- Tobin (1958)

Estimation / Concentration

- Jorion (1986); Black and Litterman (1991)
- Ledoit and Wolf (2003); DeMiguel, Garlappi, and Uppal (2009)
- Jagannathan and Ma (2003)

Risk Budgeting

- Maillard, Roncalli, and Teïletche (2010); Roncalli (2013)
- Freitas Paulo da Costa, Pesenti, and Targino (2023); Cetingoz, Fermanian, and Guéant (2024)

Flight Plan

1. MV and RB: a quick overview;
2. Our methodology: theory and simulation results;
3. Empirical application: U.S. equity market using CRSP data;
4. Some possible extensions: we **need** your input!

Empirical takeaway:

- Our methodology indeed delivers portfolios that are less concentrated than MV;
- Higher expected returns than RB;
- No way around it: moment estimation is still key;

MV and RB

Setup

- We trade d assets indexed by $i = 1, \ldots, d$;
- Returns r_i have an expected value $\mu_{d \times 1}$ and covariance matrix $\Sigma_{d \times d}$;
- An allocation $\boldsymbol{v} = (v_1, \ldots, v_d)^\intercal$ is a vector of *dollars* invested in each asset;
- You have $v_0 > 0$ dollars to invest;
- The dollar return for an allocation $\boldsymbol{v}$ is given by:

$$R(\boldsymbol{v}) \equiv \sum_{i=1}^{d} v_i \cdot r_i = v_0 \left[\sum_{i=1}^{d} w_i \cdot r_i \right], \qquad w_i \equiv \frac{v_i}{v_0} \tag{1}$$

The MV way

- You want to minimize the variance of returns $\sigma(R(\boldsymbol{v}))$;
- But you request a minimum expected return $\mu_{\min}^{MV}$;
- The (long-only) MV allocation $\boldsymbol{v}^{MV}$ solves:

$$\begin{aligned} \min_{\boldsymbol{v}\geq 0} \quad & \sigma(R(\boldsymbol{v})) \\ \text{s.t.} \quad & \textstyle\sum_{i=1}^{d} v_i = v_0 \\ & \mu(R(\boldsymbol{v})) \geq \mu_{\min}^{MV} \cdot v_0 \end{aligned} \tag{2}$$

- The MV portfolio is given by $\boldsymbol{w}^{MV} \equiv \frac{1}{v_0} \cdot \boldsymbol{v}^{MV}$;
- This is equivalent to maximizing the return given an upper bound on volatility;

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- This is equivalent to maximizing the return given an upper bound on volatility;

Drawback: MV portfolios can be highly concentrated on a few assets!

Risk Contributions

- If we increase v_i by \$1, how much does the portfolio risk $\sigma(R(\boldsymbol{v}))$ change?

Definition

The *risk contribution* of asset i to the total portfolio risk $\sigma\left(R(\boldsymbol{v})\right)$, is given by:

$$\mathcal{RC}_i(\boldsymbol{v}) \equiv v_i \cdot \frac{\partial \sigma(R(\boldsymbol{v}))}{\partial v_i}$$

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- Since $\sigma(.)$ is homogeneous of degree 1, Euler's theorem implies:

$$\sigma(R(\boldsymbol{v})) = \sum_{i=1}^{d} \mathcal{RC}_i(\boldsymbol{v}).$$

Risk Budgeting

- Risk budgeting (RB) is a framework to limit the risk contributions $\mathcal{RC}_i(\boldsymbol{v})$;
- We require a *risk budget* $\boldsymbol{b} = (b_1, \ldots, b_d)^\intercal$ for total risk and want to invest v_0;

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- No clue from the definition how to *compute* $\boldsymbol{v}$!
- The system in (3) is a non-linear system of d equations;

How to find this portfolio?

Proposition

Given a risk budget $\boldsymbol{b}$, any optimal solution $\boldsymbol{v}^*$ to

$$\min_{\boldsymbol{v}\in\mathbb{R}^d_+} \sigma(R(\boldsymbol{v}))\,, \qquad \textit{subject to} \quad \sum_{i=1}^{d} b_i \cdot \log(v_i) \geq 0 \tag{4}$$

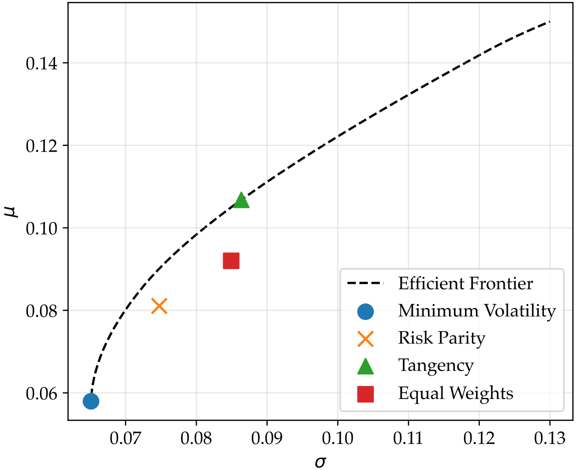
is proportional to the exposure $\boldsymbol{v}$ of the RB portfolio for risk budget $\boldsymbol{b}$.

- This problem is strictly convex and easy to solve;
- The FOC's coincide with the RB conditions in (3);
- We can rescale the solution:

$$\boldsymbol{v}^{RB} \equiv \frac{v_0}{\sum_i^d v_i^*}\boldsymbol{v}^*$$

Calibrated Example

- Let $d = 5$, and $\boldsymbol{b} = (0.2, 0.2, 0.2, 0.2, 0.2)$ – which denotes *risk parity*;
- With population moments, we compute several portfolios:



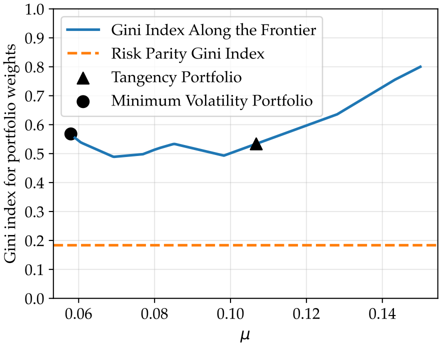
- If all you care is Sharpe Ratio, stick to MV;
- Risk parity $\neq$ Equal weights;
- You almost always have to visit the interior to get risk budgeting;

Calibration

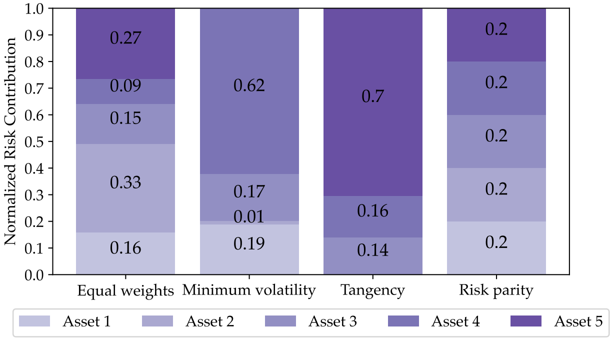
Calibrated Example: Concentration Through the Gini Index

- Given a vector $\boldsymbol{x} = (x_1, \dots, x_n)$, we have $\text{Gini}(\boldsymbol{x}) \equiv \frac{\sum_{i=1}^{n} \sum_{j=1}^{n} |x_i - x_j|}{2n \sum_{i=1}^{n} x_i}$.
- $0 \leq \text{Gini}(\boldsymbol{x}) \leq 1$, and $\text{Gini}(\boldsymbol{x}) = 0$ if and only if $\boldsymbol{x}$ is uniform;
- Brazilian (wealth) Gini: 0.52; South African Gini: 0.63; Swedish Gini: 0.29;

(a) Portfolio weight Gini indices



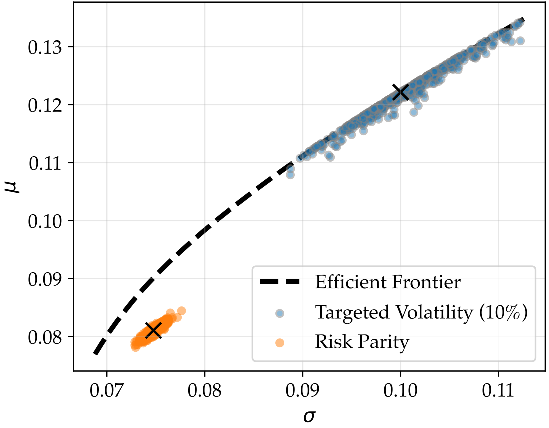
(b) Risk Contributions



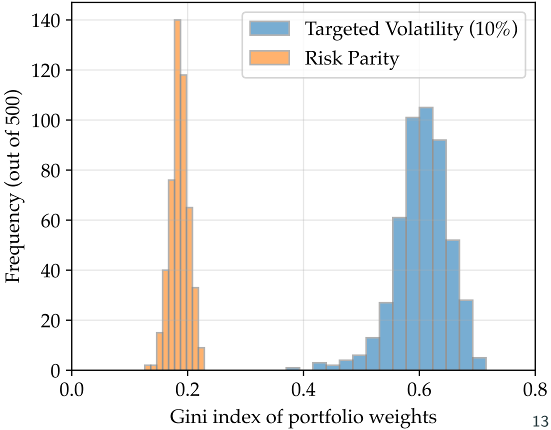
Calibrated Example: Adding Estimation Error

- Simulate 1 year of daily returns, estimate moments, and compute portfolios;
- Plot expected returns and vols using the *population moments*;

(a) Simulated portfolios in the (σ, μ)-plane



(b) Distribution of realized Gini indices for $\boldsymbol{w}_i$



Methodology

Our Methodology

Definition (The Risk-Budgeted Mean-Variance Portfolio)

Given a risk budget $\boldsymbol{b}$, an endowment v_0, a minimum required expected return μ_{min}, and a maximum volatility bound σ_{max}, the Risk Budgeted Mean-Variance Portfolio (RBMV) is given by $\boldsymbol{v} = \frac{v_0}{\sum_{i=1}^d v_i^*} \cdot \boldsymbol{v}^*$, where $\boldsymbol{v}^*$ is the solution of:

$$
\begin{aligned}
\min_{\boldsymbol{v} \in \mathbb{R}_+^d} \quad & \sigma(R(\boldsymbol{v})) & & (5) \\
\text{s.t.} \quad & \sum_{i=1}^d b_i \log(v_i) \geq 0 & [\lambda_v] & \\
& \mu(R(\boldsymbol{v})) \geq \mu_{\text{min}} \textstyle\sum_{i=1}^d v_i & [\lambda_\mu] & \\
& \sigma(R(\boldsymbol{v})) \leq \sigma_{\text{max}} \textstyle\sum_{i=1}^d v_i \,, & [\lambda_\sigma] &
\end{aligned}
$$

The corresponding portfolio weights are given by $\boldsymbol{w} = \frac{1}{v_0} \boldsymbol{v}$.

We adapt this from Maillard, Roncalli, and Teïletche (2010) and Roncalli (2014).

Our Methodology: The Lagrangian View

$$\frac{\partial \mathcal{L}(\boldsymbol{v}; \lambda_v, \lambda_\mu, \lambda_\sigma)}{\partial v_i} = \frac{\partial \sigma(R(\boldsymbol{v}))}{\partial v_i} - \lambda_v \frac{b_i}{v_i} - \lambda_\mu \left(\mu_i - \underbrace{\mu_{\text{min}}}_{\color{red}{\text{moves around}}} \right) - \lambda_\sigma \left(\underbrace{\sigma_{\text{max}}}_{=0.1} - \frac{\partial \sigma(R(\boldsymbol{v}))}{\partial v_i} \right)$$

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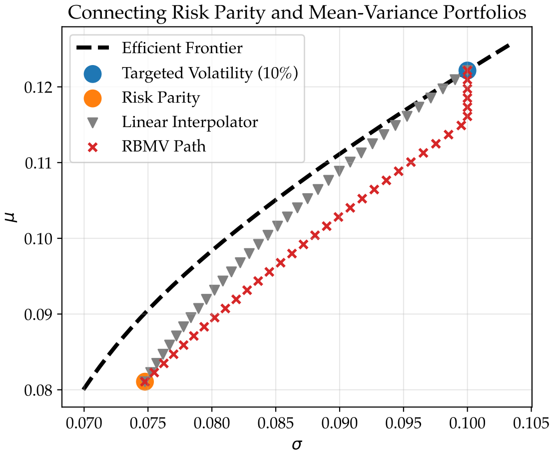

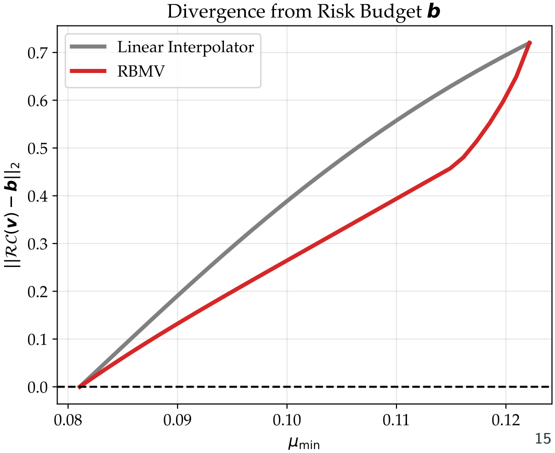

Empirical Application

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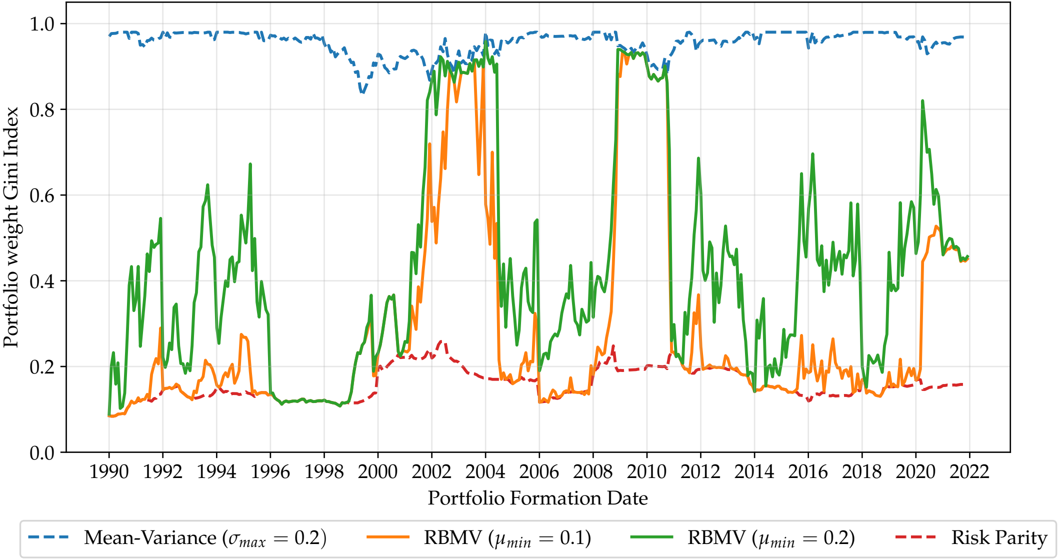
- Every month, we build a long-only portfolio with the 50 largest stocks in the U.S.;
- Sample: 1990-2023;
- Estimate population moments $\hat{\mu}$ and $\hat{\Sigma}$ with 2 years of historical data (CRSP);
- Solve for the minimum vol portfolio and set $\sigma_{max} = \min\{\sigma_{MinVol} + 0.02, 0.2\}$;
- Find the MV portfolio with the highest returns, given σ_{max};
- Set two possible values for μ_{min}:

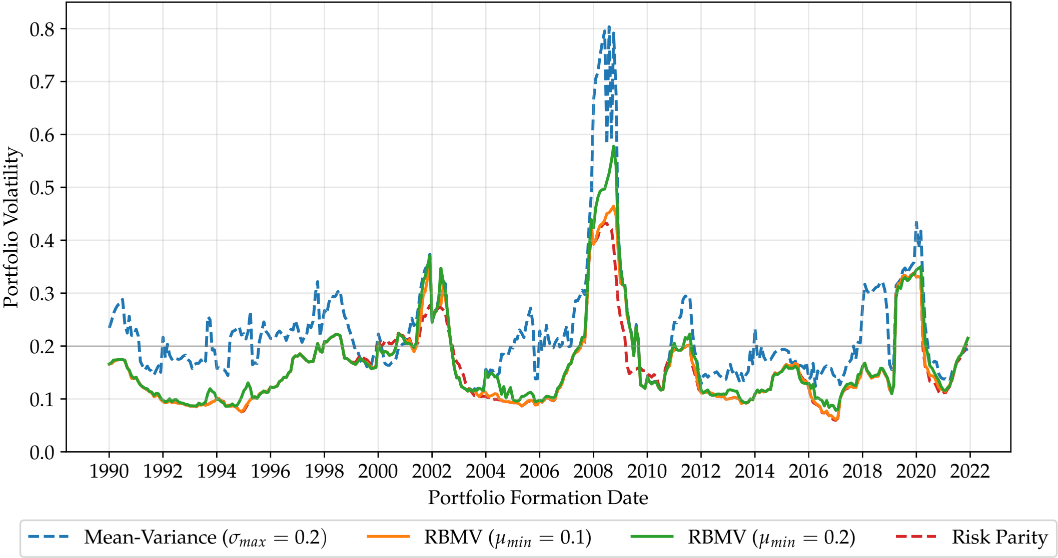
$$\mu_{min,\text{ conservative}} \equiv \min\{\mu_{MV} - 0.05, 0.1\}$$
$$\mu_{min,\text{ greedy}} \equiv \min\{\mu_{MV} - 0.05, 0.2\}$$

- Hold 4 portfolios for 1 year: Risk Parity, MV(σ_{max}), RBMV($\mu_{min,\text{conservative}}; \sigma_{max}$), and RBMV($\mu_{min,\text{greedy}}; \sigma_{max}$);

Realized Gini Index



Annualized Standard Deviation of Daily Returns



Portfolios in the (σ, μ)-Plane

Average results

Full Sample (1990-2022)

Portfolio	**Return (%)**	**Volatility (%)**	**Sharpe Ratio**	**Gini Index (w_i)**
Risk Parity	9.68	15.69	0.84	0.16
RBMV ($\mu_{min} = 0.1$, $\sigma_{max} = 0.2$)	10.29	15.98	0.86	0.29
RBMV ($\mu_{min} = 0.2$, $\sigma_{max} = 0.2$)	10.50	16.70	0.81	0.43
Mean-Variance ($\sigma_{max} = 0.2$)	10.20	22.28	0.58	0.96

- We were able to tilt the Risk Parity portfolio towards higher returns;
- Just a bit more of concentration led to higher returns and slightly higher SR;
- The MV portfolio was invested in just a few assets!

▸ Alternative Samples ▸ Realized Sharpe Ratio ▸ Realized Returns ▸ σ_{max} ▸ μ^{MV}

Discussion and Extensions

Risk Measures Beyond Volatility

- Let $\rho(\boldsymbol{v}) \equiv \rho(R(\boldsymbol{v}))$ be a portfolio risk measure;
- The key property for risk budgeting is positive homogeneity;

$$\rho(\gamma \boldsymbol{v}) = \gamma \rho(\boldsymbol{v}), \qquad \gamma > 0.$$

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- Prime example: Expected Shortfall on losses $L(\boldsymbol{v}) = -R(\boldsymbol{v})$;
$$\mathrm{ES}_\alpha(L(\boldsymbol{v})) = \mathbb{E}\left[L(\boldsymbol{v}) \mid L(\boldsymbol{v}) \geq \mathrm{VaR}_\alpha(L(\boldsymbol{v}))\right].$$

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$$\mathrm{ES}_\alpha(L(\boldsymbol{v})) = \mathbb{E}\left[L(\boldsymbol{v}) \mid L(\boldsymbol{v}) \geq \mathrm{VaR}_\alpha(L(\boldsymbol{v}))\right].$$
- For any ρ with a reasonable concept of gradient, define risk contributions $\mathcal{RC}_i^\rho(\boldsymbol{v}) \equiv v_i \frac{\partial \rho(\boldsymbol{v})}{\partial v_i}$;
- Euler's theorem gives the same risk-decomposition logic:
$$\rho(\boldsymbol{v}) = \sum_{i=1}^{d} \mathcal{RC}_i^\rho(\boldsymbol{v}).$$

Short Sales: What Breaks?

- Long-only RB/RBMV uses positive dollar exposures:

$$\mathcal{RC}_i^{\rho}(\boldsymbol{v}) \equiv v_i \frac{\partial \rho(\boldsymbol{v})}{\partial v_i}, \qquad b_i \rho(\boldsymbol{v}) = \mathcal{RC}_i^{\rho}(\boldsymbol{v})$$

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- If $v_i < 0$, then $\log(v_i)$ is undefined;
- $\mathcal{RC}_i^{\rho}(\boldsymbol{v})$ can be negative, so it is no longer a positive share of total risk;
- $\sum_i v_i = v_0$ becomes net exposure, not the size of the long/short book.

We cannot just remove the constraint $\boldsymbol{v} \geq 0$.

Short Sales: Fix Signs

- Let the investor fix the signs before optimization:

$$s_i \in \{-1, +1\}, \qquad x_i > 0, \qquad v_i = s_i x_i.$$

- $s_i = -1$ means asset i is chosen in advance to be short;
- Budget risk over position magnitudes $\boldsymbol{x}$:

$$R_s(\boldsymbol{x}) \equiv R(\boldsymbol{s} \odot \boldsymbol{x}) = \sum_{i=1}^{d} s_i x_i r_i,$$
$$\widetilde{\mathcal{RC}}_i(\boldsymbol{x}) \equiv x_i \frac{\partial \rho(R_s(\boldsymbol{x}))}{\partial x_i}.$$

Short Sales: Budget Magnitudes

- Once signs are fixed, homogeneity works in the magnitudes:

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- The log formulation is also well-defined because $x_i > 0$:

$$\min_{\boldsymbol{x} \in \mathbb{R}^d_{++}} \quad \rho(R_s(\boldsymbol{x})) \quad \text{s.t.} \quad \sum_{i=1}^{d} b_i \log(x_i) \geq 0.$$

Easy once signs are fixed; hard if the model must choose what to short.

Decision-Theoretic Foundations: The MV Case

- CARA utility: $U(W) = -\exp(-\gamma W)$, $\gamma > 0$; Returns: $R(\boldsymbol{v}) \sim \mathcal{N}(\mu(R(\boldsymbol{v})), \sigma^2(R(\boldsymbol{v})))$;

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$$\max_{\boldsymbol{v}} \quad \mu(R(\boldsymbol{v})) - \frac{\gamma}{2}\sigma^2(R(\boldsymbol{v}));$$

- This is exactly a mean-variance criterion: every solution lies on Markowitz's frontier, γ pins down which point.
- Gaussianity collapses the distribution to (μ, σ^2); CARA lets v_0 factor out.

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  - The RB portfolio is defined by the system $b_i \cdot \sigma(R(\boldsymbol{v})) = \mathcal{RC}_i(\boldsymbol{v})$;
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- A natural conjecture: the risk budget $\boldsymbol{b}$ acts as a *reference point* for risk allocation
  - Reminiscent of preference models where agents evaluate outcomes relative to a benchmark;
  - Can something like that rationalize the RB conditions? Not sure, but I would like to find out!

Conclusion

Wrap-Up:

- MV is the cornerstone of portfolio allocation, but leads to high concentration;
- RB tackles concentration at the expense of expected returns;
- RBMV nests both, and allows you to control this trade-off in a transparent way;
- A Julia package for fast implementation: https://github.com/bfpc/RiskBudgetingMeanVariance.jl

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Going forward:

- Some extensions: short-selling, transaction costs, ...;
- What kind of other markets can benefit from this methodology? What examples are interesting?

Appendix and References

Calibration Details

We assume $d = 5$ and use the following population moments for the mean returns μ, the individual standard deviations s and correlation matrix C:

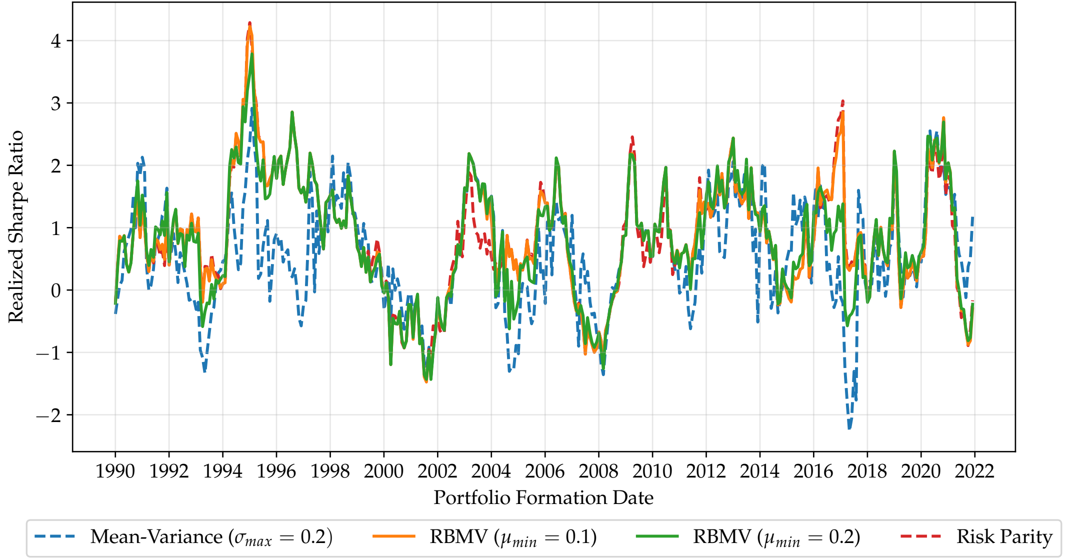
$$\mu = \begin{bmatrix} 0.5 \\ 0.12 \\ 0.09 \\ 0.05 \\ 0.15 \end{bmatrix}, \quad s = \begin{bmatrix} 0.10 \\ 0.20 \\ 0.15 \\ 0.08 \\ 0.13 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0.20 & 0.40 & 0.25 & 0.50 \\ 0.20 & 1 & -0.20 & 0.40 & 0.6 \\ 0.40 & -0.20 & 1 & -0.10 & 0.30 \\ 0.25 & 0.40 & -0.10 & 1 & 0.30 \\ 0.50 & 0.60 & 0.30 & 0.30 & 1 \end{bmatrix} \tag{6}$$

We further define $\Sigma \equiv s^{\mathsf{T}} C s$.

▸ Back to Example

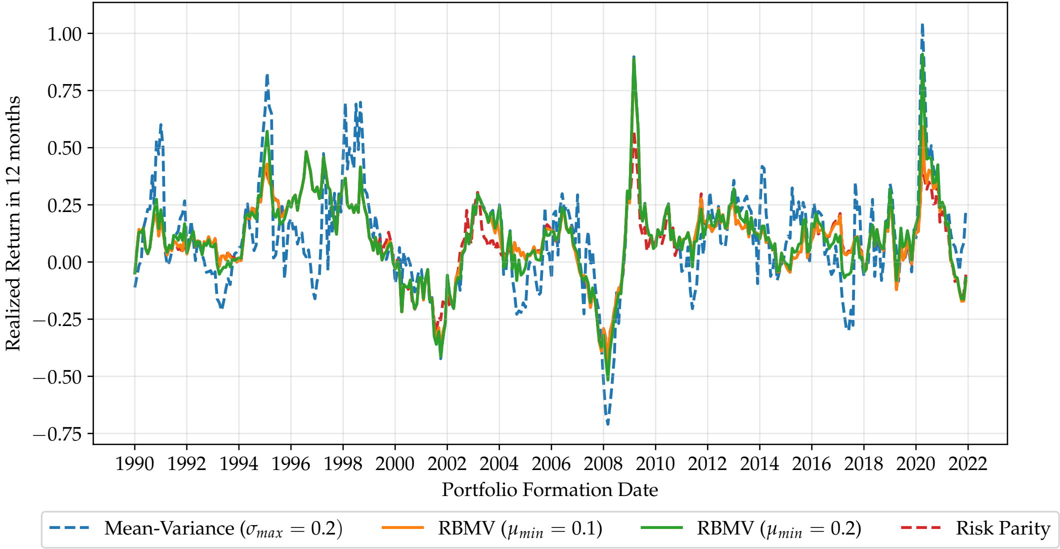
Realized Sharpe Ratio

▸ Back to Average Results



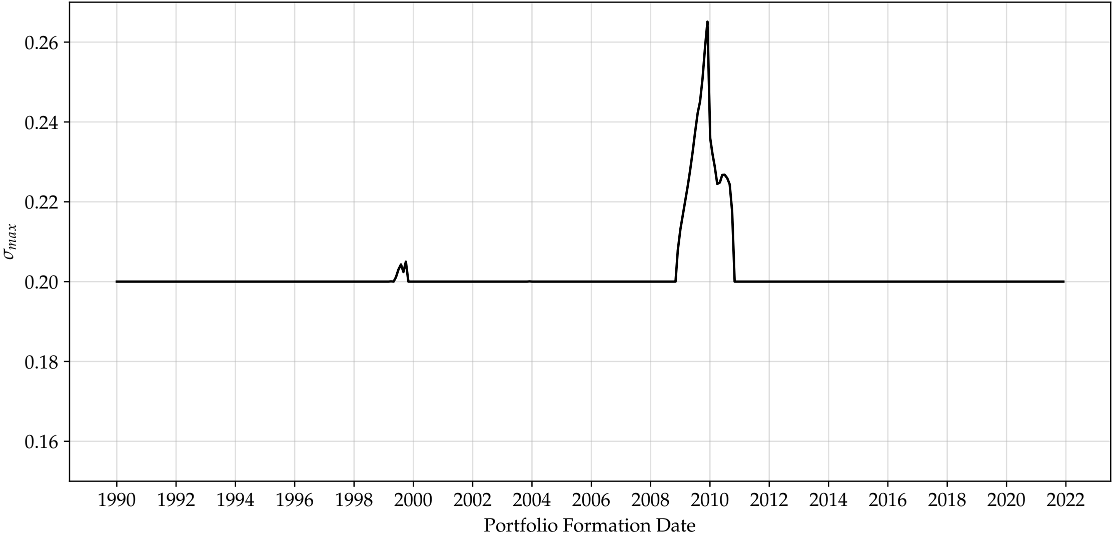
Realized Returns

▸ Back to Average Results



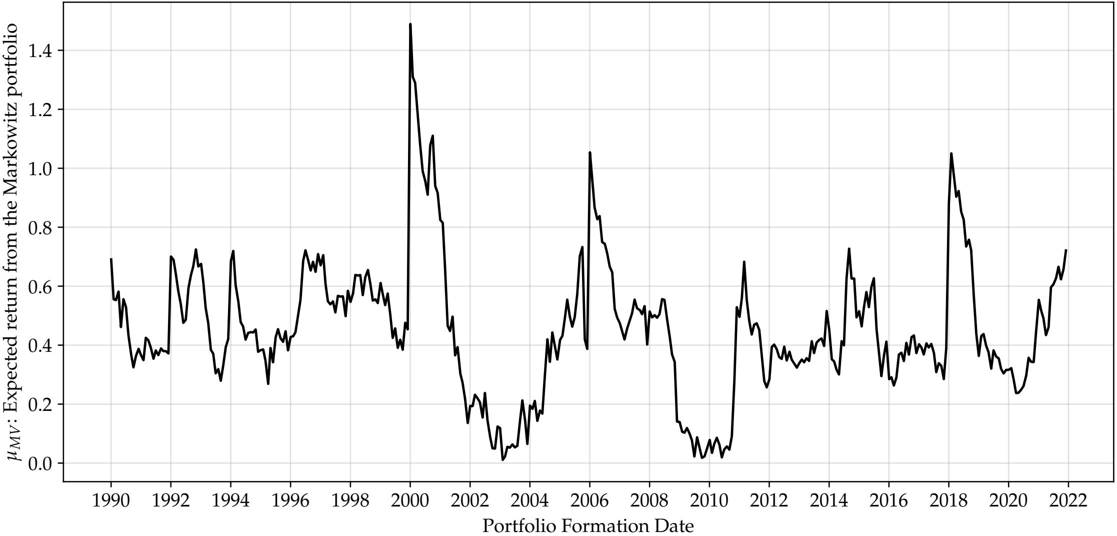
Actual Maximal Volatility

Back to Average Results



Expected Return From The MV Approach

Back to Average Results



Alternative Samples

Back to Average Results

Panel B: Before the Great Financial Crisis (1990-2006)

Portfolio	Return (%)	Volatility (%)	Sharpe Ratio	Gini Index (w_i)
Risk Parity	10.79	14.42	0.89	0.15
RBMV ($\mu_{min} = 0.1$, $\sigma_{max} = 0.2$)	10.90	14.45	0.91	0.26
RBMV ($\mu_{min} = 0.2$, $\sigma_{max} = 0.2$)	10.74	14.99	0.85	0.39
Mean-Variance ($\sigma_{max} = 0.2$)	9.92	20.84	0.50	0.95

Panel C: After 2020 (2021-2022)

Portfolio	Return (%)	Volatility (%)	Sharpe Ratio	Gini Index (w_i)
Risk Parity	0.07	15.65	0.21	0.16
RBMV ($\mu_{min} = 0.1$, $\sigma_{max} = 0.2$)	0.93	16.00	0.27	0.46
RBMV ($\mu_{min} = 0.2$, $\sigma_{max} = 0.2$)	1.52	16.01	0.30	0.47
Mean-Variance ($\sigma_{max} = 0.2$)	12.62	16.31	0.84	0.96

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